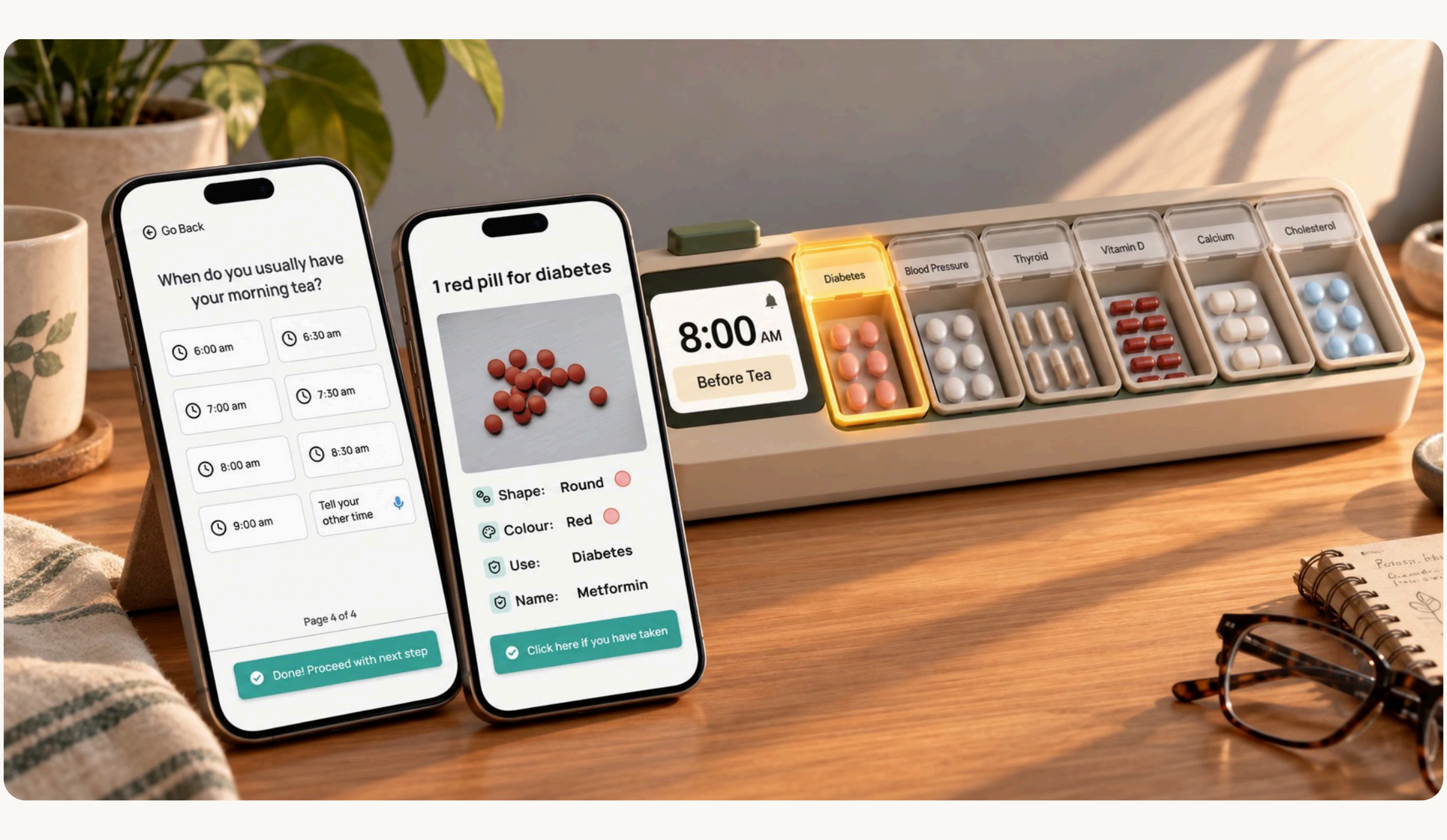
Designing Grouped Reminders to Reduce Cognitive Load in Elderly Polypharmacy Management

My Role
Ux Research, Use interviews, Behavioural Research, Interaction design, Information architecture, Visual design.

Project Type
UX Research, Service design, application design, Product design

Tools Used



Introduction

Managing multiple medicines is a daily challenge for many older adults, especially when prescriptions change and family can't always be around. This case study explores a semester-long **UX project on simplifying polypharmacy management for older adults** and their families.

In my research with 7 older adults, 9 family members and 102 survey respondents, I found that existing digital tools are often difficult for older adults to use. **I designed grouped medicine reminders that aim to reduce cognitive load**, built around detailed daily routines, even small ones like "before tea", with a connected caregiver app for families.

This project was part of my college coursework and is my attempt to make everyday healthcare a little easier.

What does the current scenario look like?

1. **Up to 12 medicines a day, tracked from memory**
   70% of those surveyed still rely on memory to take them.
2. **Medical records kept in too many places**
   65% store them physically, so quick access is hard.
3. **Existing reminders run on the clock, not on routines**
   Most apps rely on fixed timings, with no support for small routines like "before tea".
4. **Pills sorted by hand, every week**
   Filling day and night compartments is repetitive and time-consuming.

The solution

Two connected digital products
one designed for **elderly users** to manage their medicines, and another for **family caregivers** to support and manage their care.

Medicines grouped by timing and routine
Medicines taken together sit in one group, so a single reminder covers the whole routine.

A routine check before the reminder
A pop-up first asks "Did you have your breakfast?", then reminds them at the right moment.

Add reminders by voice or selector chips
Say it or tap a chip to set up small routines like "before tea".

Medicine remainders easy to recognise

Family medicine tracking
Caregivers see which medicines were taken or missed, at a glance.

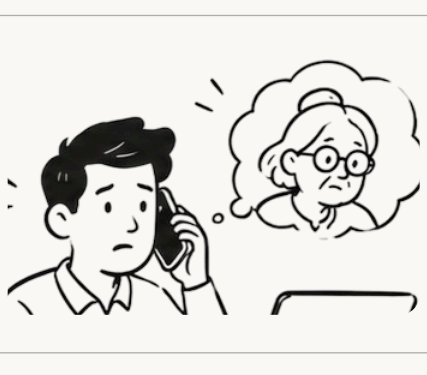
Book caregiver for specific needs

Qualitative Research Insights

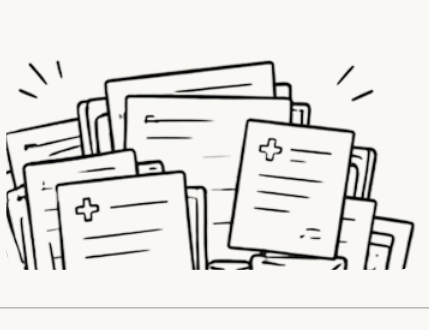
Interviewed 7 elderly people, 9 family members, 1 medical student

"When my routine changes, I forget my medicines."

"I remember medicines through routines, like after brushing."

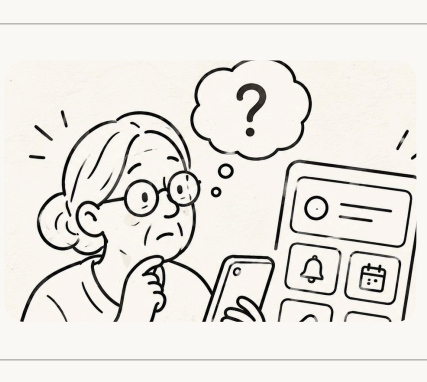


"I do not want to depend on my family all the time"



" Medical records are scattered"

"We need a caregiver, but it's hard to find one."



"I want to use digital tools, but they're hard to use."

Quantitative Research Insights

Surveyed 102 people

70%
"Still rely on memory"

65%
"Store medical records phscially"

52%
"See value in medicine remainders"

Secondary Research Insights

Older adults recognise medicines by sight
Shape, size, colour and where they keep them.

Medicines are tied to routines, not clock times
"Before tea", "after lunch".

Reminders matter most when a medicine is new
They help until the medicine becomes a habit.

User personas

01 ELDERLY PERSON

Meena Joshi · 68
Retired teacher · Lives with her husband

Meena takes multiple medicines every day. She has developed her own way of remembering them through their appearance and her daily routine.

Needs
- Easily recognise the right medicine
- Store medical history and records
- Add medicines and records easily
- Get reminders when needed
- Adaptable remainder for routine change

Frustrations
- Similar-looking medicines get confusing
- Changing prescriptions disturb her routine
- Too many steps or information can feel overwhelming.

02 FAMILY CAREGIVER

Rahul Joshi · 39
Working professional · Meena's son

Rahul helps manage her mothers healthcare, but work means he cannot always be around to check medicines, appointments or accompany her

Needs
- Know whether medicines are taken
- Quickly add or Update medicines
- Keep records and appointments together
- Find trustable caregiver
- Adaptable remainder for routine change

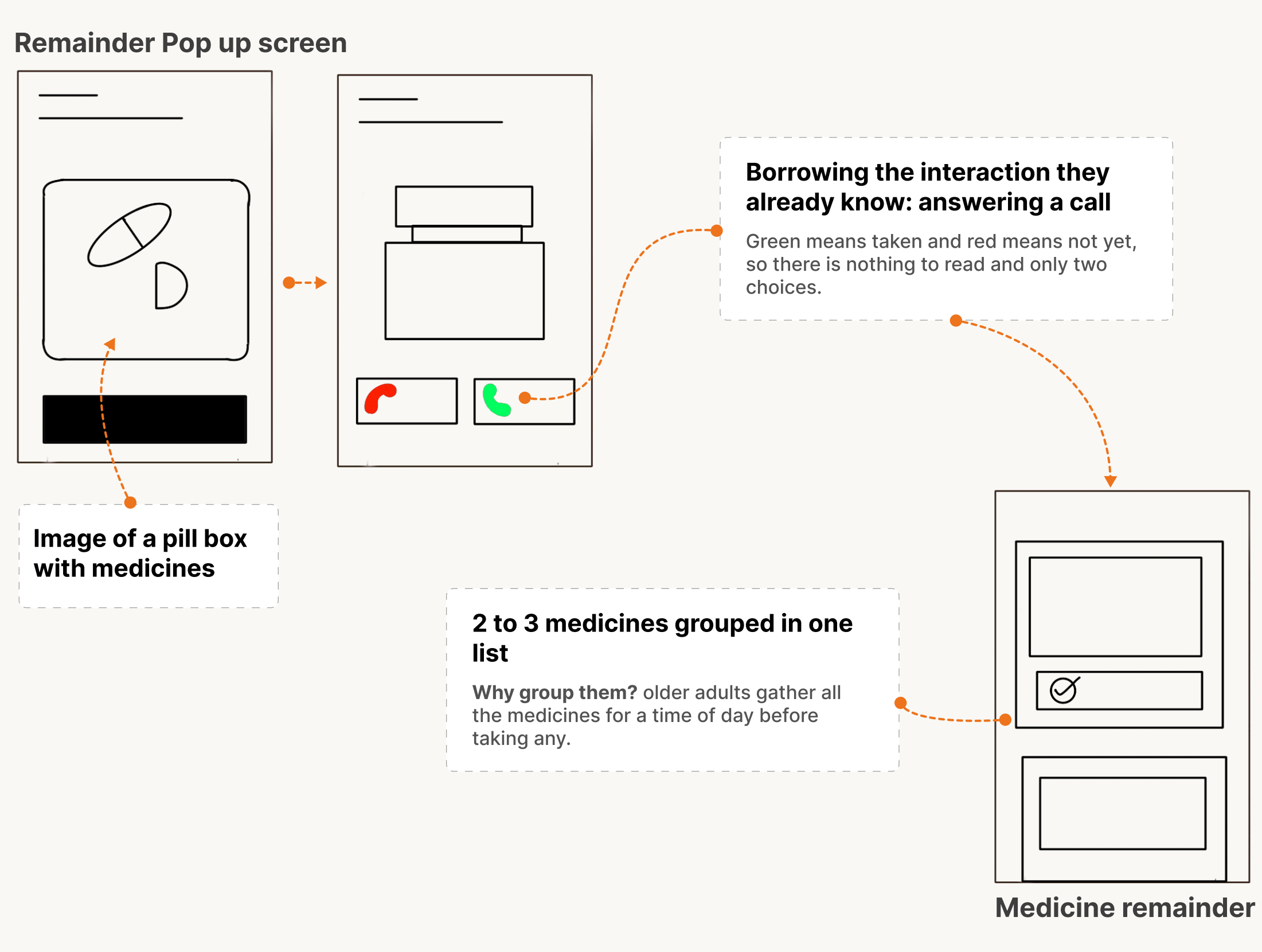
Frustrations
- Managing many medicines takes time
- Medical information is scattered
- Hard to know what happened when he wasn't there

PROBLEM STATEMENT

> *"How might we reduce the cognitive load of managing many medicines, while keeping families involved?"*

Why ask about *"Did you have your breakfast"* before asking about medicine?

Remainder Pop up screen



Borrowing the interaction they already know: answering a call
Green means taken and red means not yet, so there is nothing to read and only two choices.

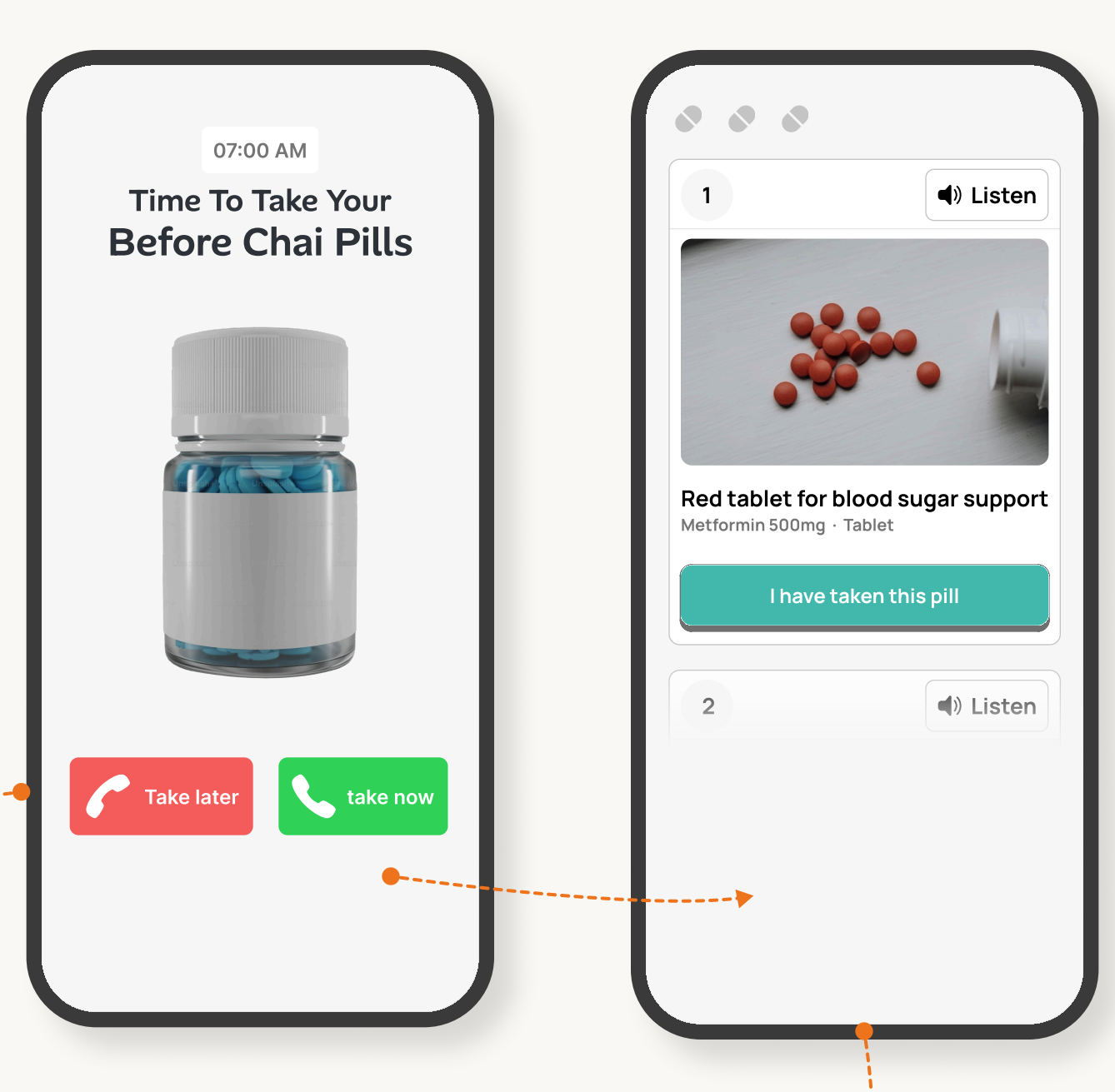
Image of a pill box with medicines

2 to 3 medicines grouped in one list
Why group them? older adults gather all the medicines for a time of day before taking any.

Medicine remainder

Iteration

Call-style reminder failed: The buttons didn't say what happens next
Proof from usability testing
- Users were confused by the call interaction.
- "Take now" and "Take later" didn't tell them what would happen after tapping.



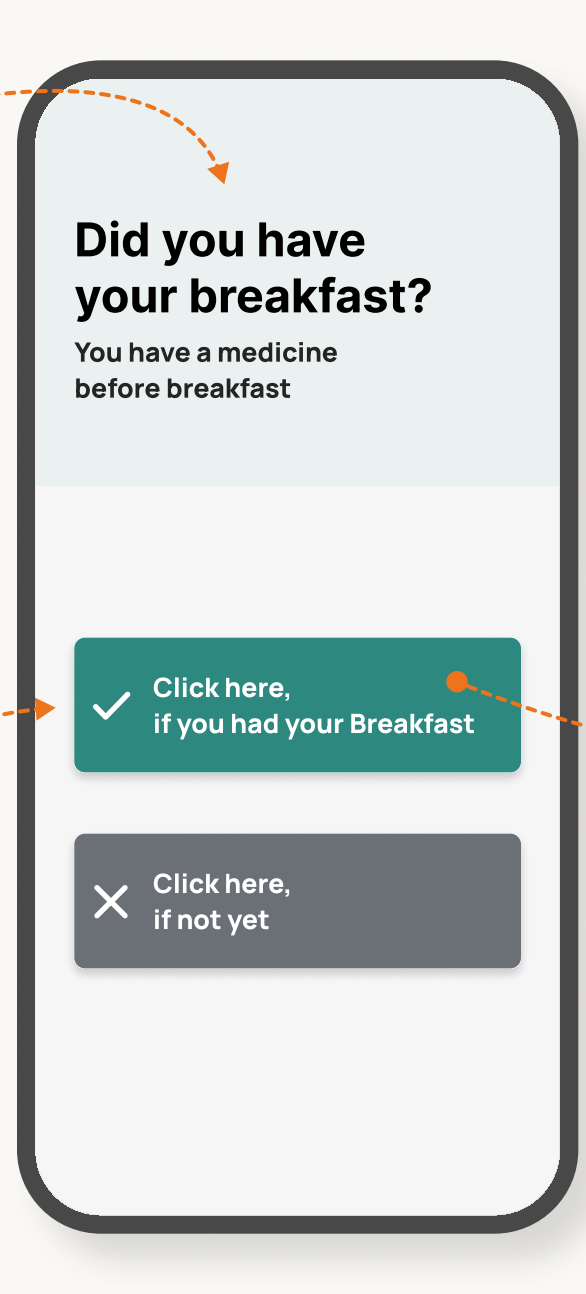
07:00 AM
Time To Take Your Before Chai Pills

Grouped medicine list failed: Because medicines are taken 15 to 20 minutes apart
Proof from usability testing
- There is usually a gap of 15 to 20 minutes between them.
- Extra cards added clutter.

Final screens

Asking About the Routine Before Showing the Medicine Reminder
Testing revealed: Users mistook the earlier pop-up for the medicine reminder itself.

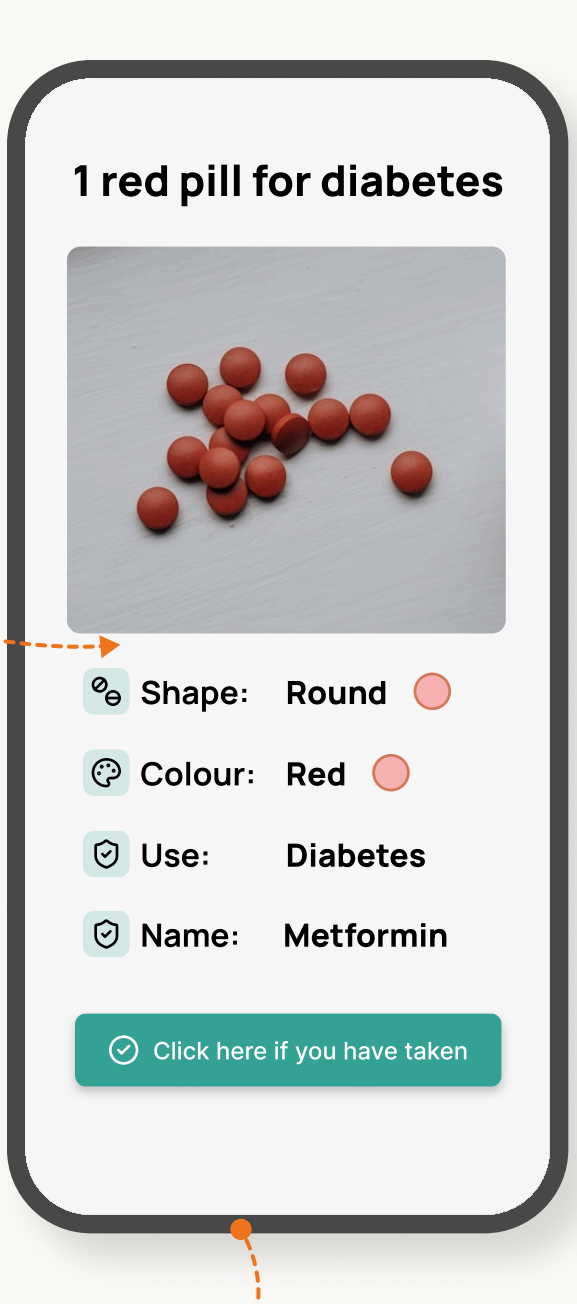
Design decision: Ask whether breakfast is completed before showing the relevant medicine reminder.



Did you have your breakfast?
You have a medicine before breakfast

Using Descriptive Buttons to Make the Next Action Clear
Design decision: Added explicit button labels that communicate the action and its outcome.

Principle Applied
Contextual prompts: Checked the routine before showing the relevant reminder.

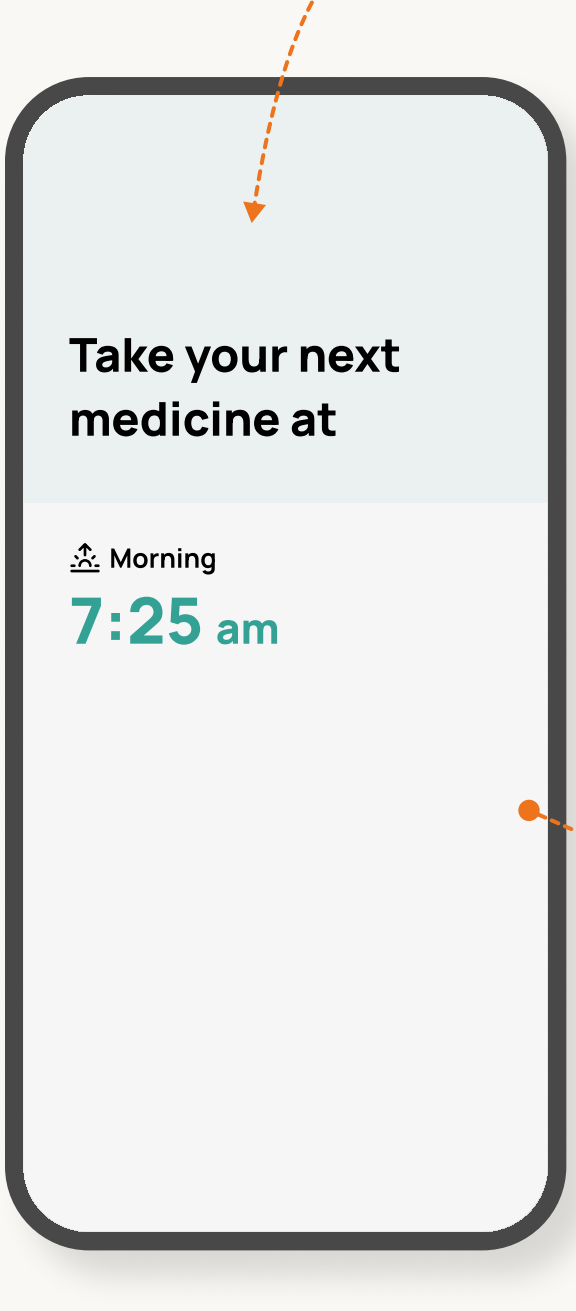


1 red pill for diabetes
Shape: Round
Colour: Red
Use: Diabetes
Name: Metformin

Adding Medicine Images, Colour, Shape and Purpose for Easier Recognition
Interviews revealed: Users remember medicines by their common name, purpose, appearance and storage location.

Added familiar visual and descriptive cues instead of relying only on clinical names.

Testing confirmed: Users recognised medicines easily.

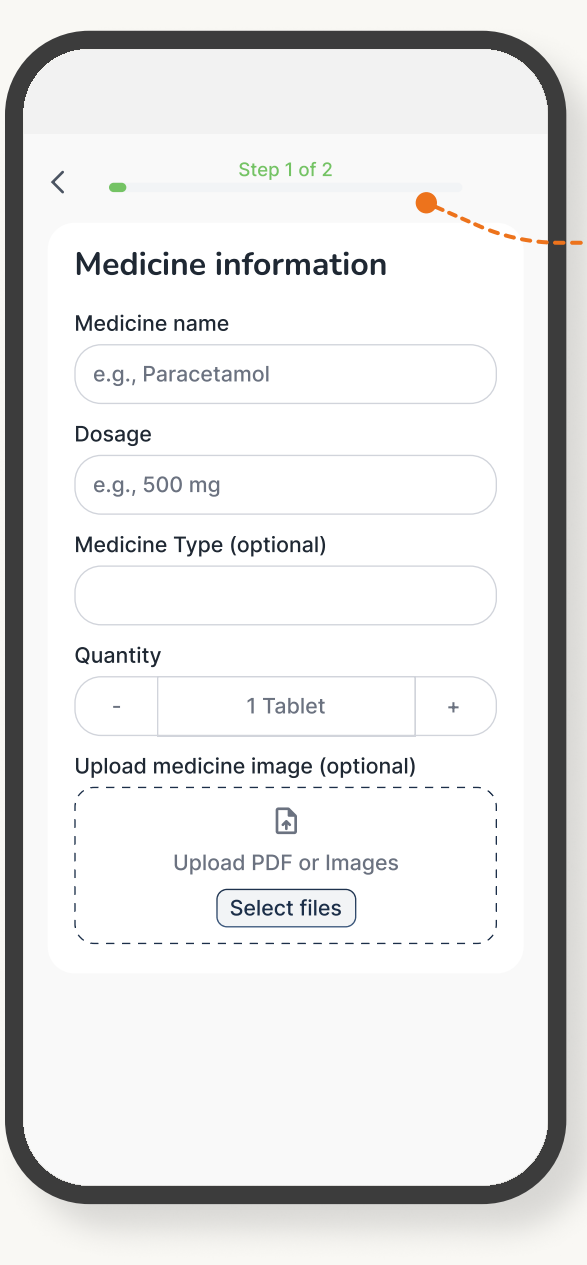


Take your next medicine at
Morning
7:25 am

Showing Only the Next Medicine Time After Each Reminder
This **maintains continuity between doses** without overwhelming users with unnecessary information.

Asking *When they have their tea?* instead of what time to set the alarm?

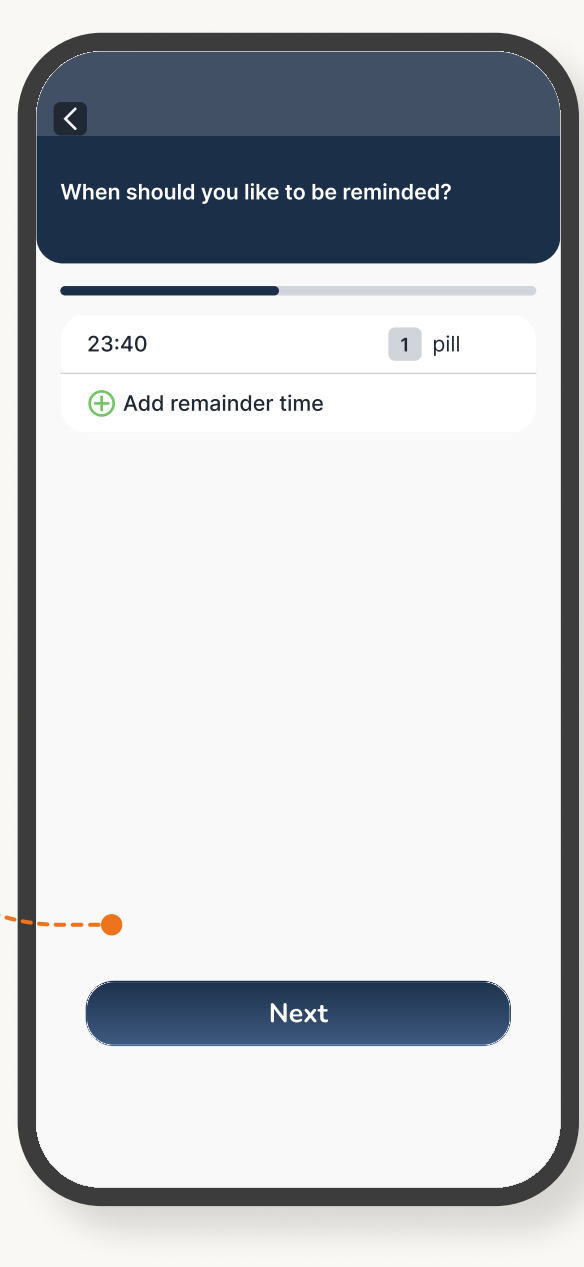
Iteration 01



Step 1 of 2
Medicine information
Medicine name: e.g., Paracetamol
Dosage: e.g., 500 mg
Medicine Type (optional)
Quantity: - 1 Tablet +
Upload medicine image (optional)
Upload PDF or Images

Too Many Medicine Details Made Adding Medicines Tedious
Testing revealed: Entering multiple medicine details at once was time consuming and overwhelming.

Iteration 02

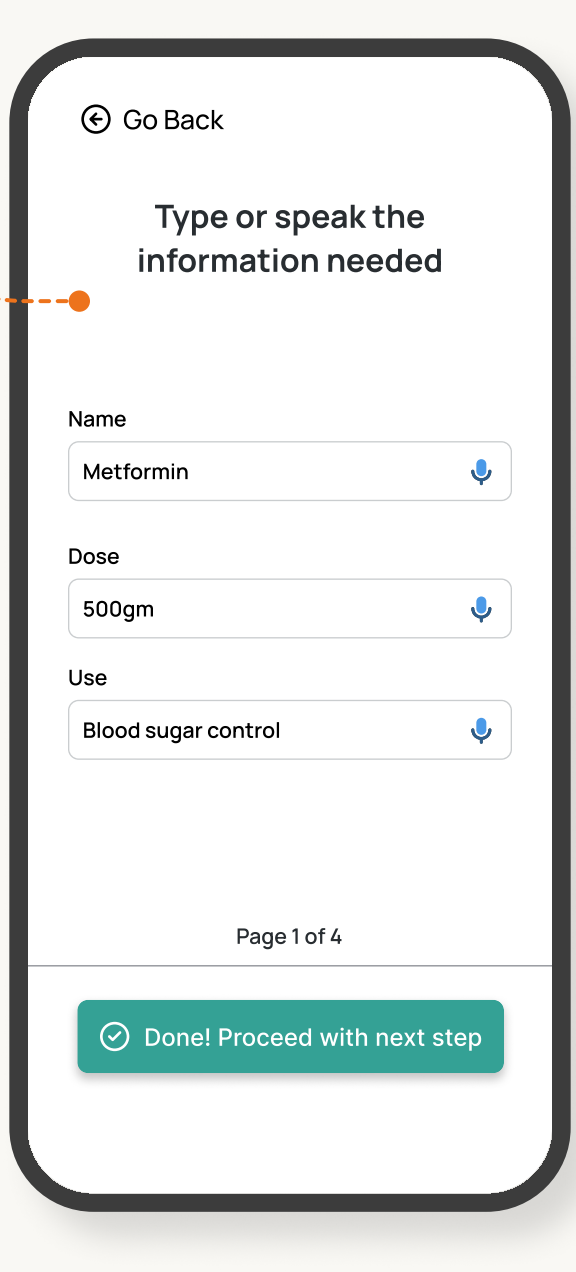


When should you like to be reminded?
23:40 1 pill

Users Remember Medicine Timings Through Daily Routines, Not Fixed Clock Times
Interviews revealed: Users remember medicines through routines like tea or breakfast, which may vary in timing.

Iteration 03

Testing revealed: Too many questions on one screen confused users.

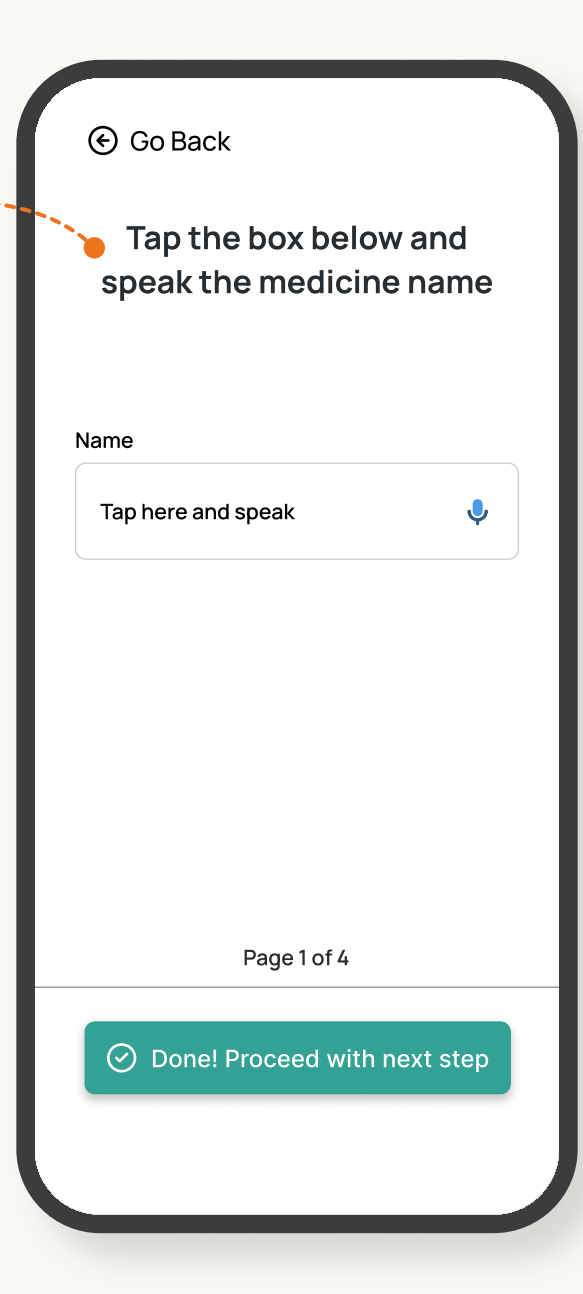


Go Back
Type or speak the information needed
Name: Metformin
Dose: 500gm
Use: Blood sugar control
Page 1 of 4

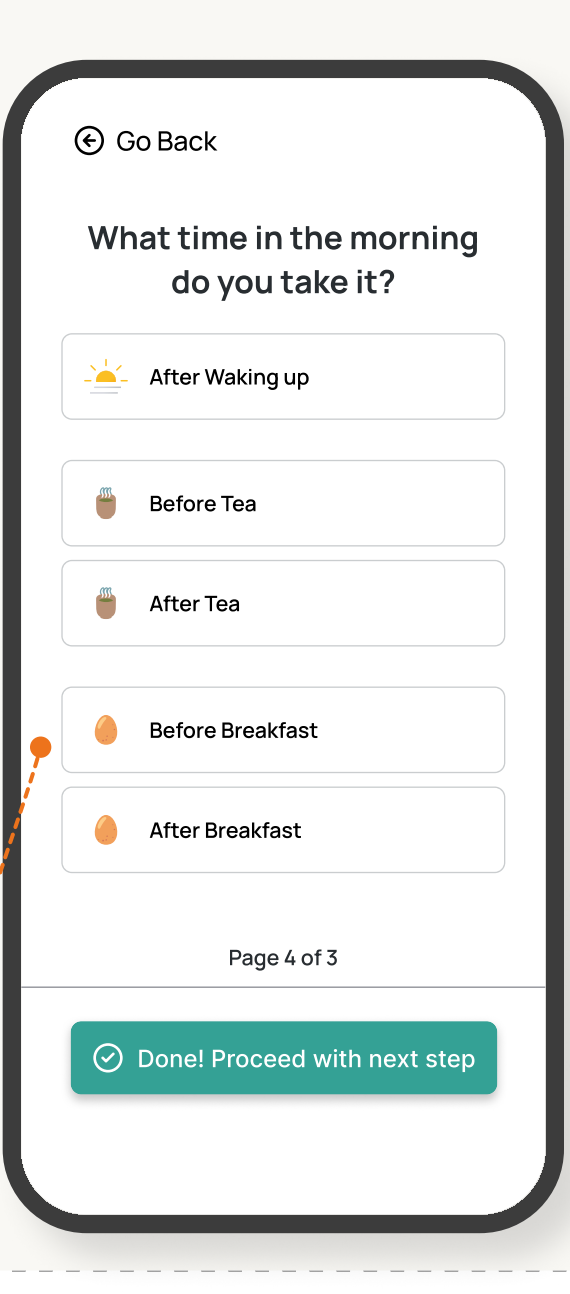
Final Flow

Asking One Question per Screen
They are more comfortable with voice commands then typing.

Principle Applied
Reduced cognitive load: Showing only what is needed at each step.



Go Back
Tap the box below and speak the medicine name
Name: Tap here and speak
Page 1 of 4



Go Back
What time in the morning do you take it?

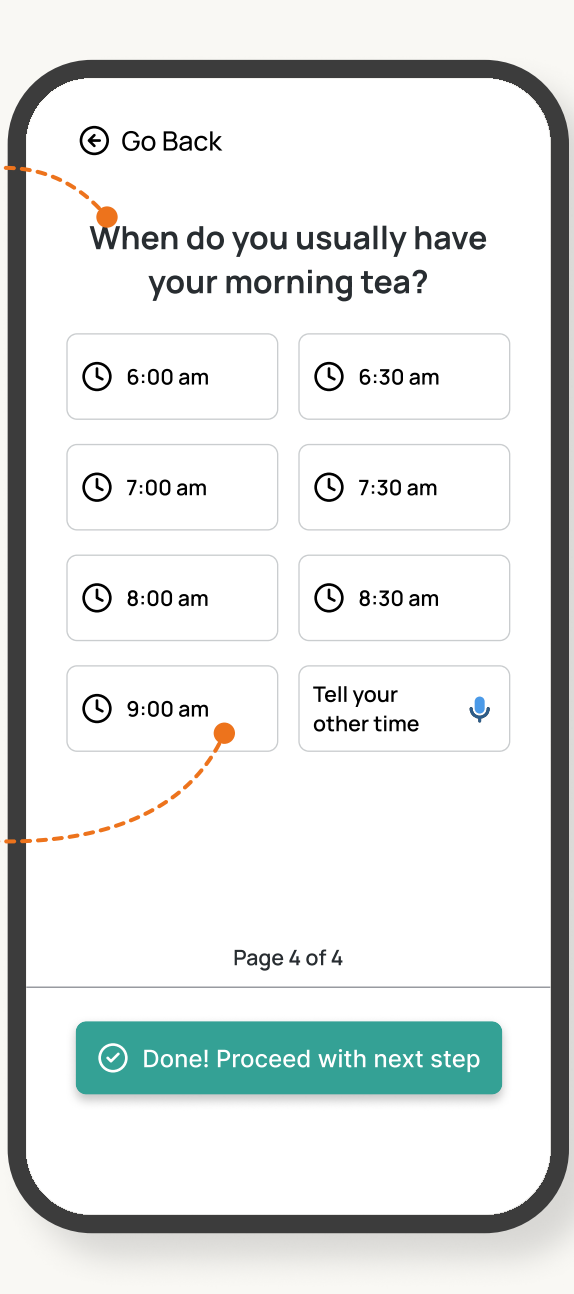
Adding Detailed Routine Options for Personalised Reminders
Interviews revealed: Users associate medicines with specific daily activities.

Added options like before/after tea and breakfast to match existing habits.

Principle Applied
Habit Anchoring: Connecting medicines to familiar routines.

Asking for Routine Timing Instead of Medicine Timing
Interviews revealed: Users recall when they have tea more easily than medicine timings.

Used routine timing to help set reminders.



Go Back
When do you usually have your morning tea?

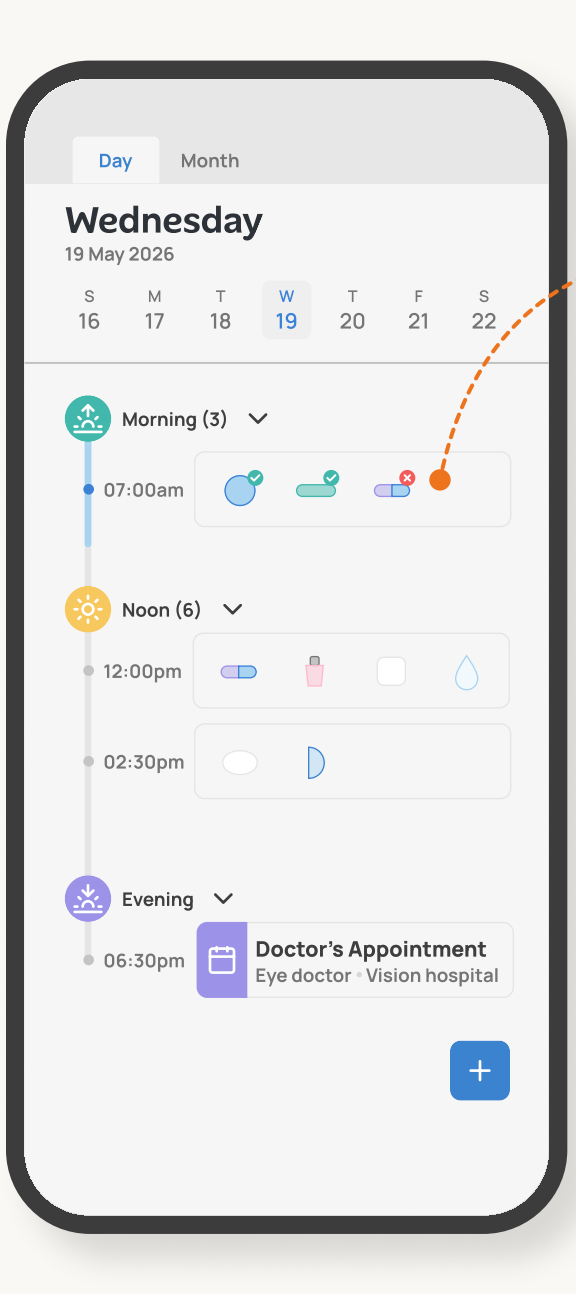
Replacing Manual Time Entry with Selectable Chips and Voice Input
Testing revealed: Users preferred selecting options or speaking over typing.

Added time chips and voice input to reduce effort.

Naming medicine list by what they treat, not their clinical names

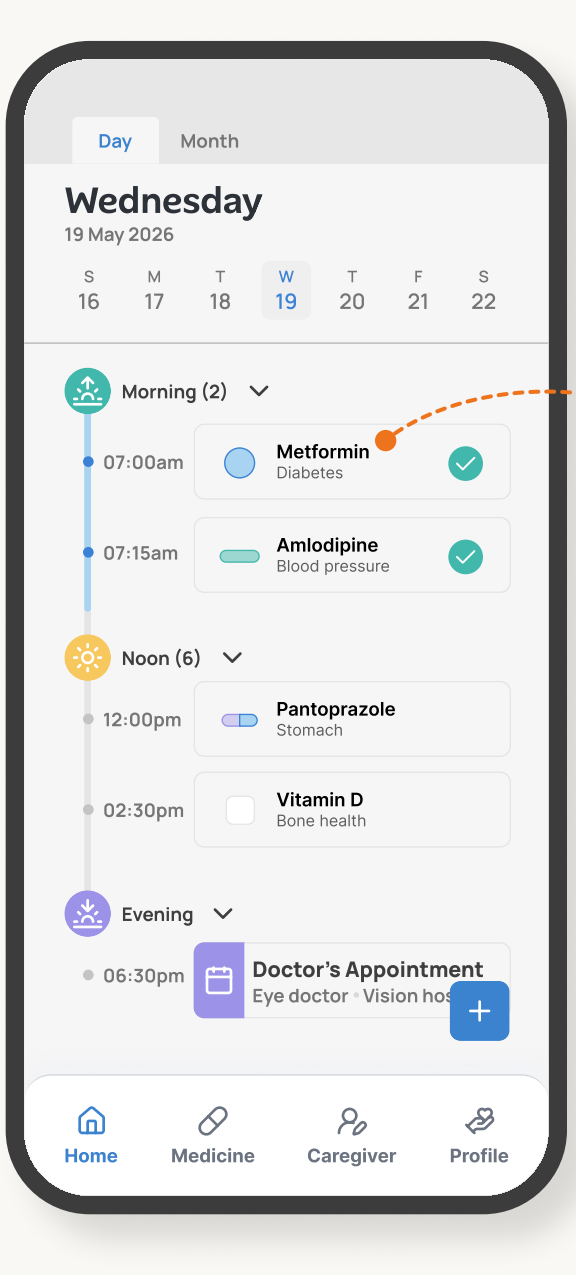
Family caregiver app

Iteration 01



Testing revealed: Users struggled to identify medicines without names.

Iteration 02



Testing revealed: Users took 1–2 minutes to understand the medicines, because the clinical names were highlighted

Final screen

Highlighting Medicine Purpose Instead of Clinical Names
Prioritised familiar labels like "Diabetes" and "Blood Pressure" for quicker identification.